NuGAT: Evidence-Grounded Multimodal Reasoning via Nash Utility-Guided Attention Transport

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Abstract

Multimodal reasoning requires models to ground task-relevant evidence from heterogeneous modalities, such as text, images, audio, video, and structural. Despite recent progress, existing methods often yield fragmented evidence grounding and rely on fixed alignment costs, predefined shared spaces, or static fusion weights. As a result, they struggle to explicitly model evidence-anchor correspondences and remain vulnerable to modality dominance, where strong modalities overwhelm weaker but complementary ones. We propose NuGAT, a general evidence-grounded multimodal alignment and fusion framework based on Attention Transport (AT) and Nash Utility Fusion (NUF). AT formulates evidence grounding as a structured evidence-anchor correspondence learning problem, in which modality-specific evidence units are adaptively transported to evidence anchors through attention-guided and structure-aware transport. This design produces explicit and interpretable evidence-anchor correspondences beyond fixed alignment costs. NUF treats each modality as a utility-bearing modality and estimates its utility gain by comparing its alignment quality with a modality-specific disagreement state. Rather than using utility gain as a scalar fusion weight, NuGAT uses it to control modality-specific subspace retention, preserving reliable evidence directions while suppressing noisy or weakly grounded signals. Experiments on multimodal QA and concept verification demonstrate the effectiveness of NuGAT. Compared with the recent strongest baseline, NuGAT achieves an average improvement of 3.37 across audio-visual QA under fixed answerers.

Keywords: Multimodal Grounded Reasoning; Attention Transport; Nash Utility Fusion.

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1. Introduction

Multimodal reasoning requires models to ground task-relevant evidence from heterogeneous modalities, such as text, images, audio, video, and structural [1, 2]. Although recent Multimodal Large Language Models (MLLMs) have demonstrated strong perception and generation capabilities [3, 4], complex tasks still require fine-grained evidence grounding beyond coarse multimodal understanding. When answers depend on cross-modal evidence, models must identify concrete evidence units rather than rely on the entire multimodal context [5, 6, 7, 8]. Therefore, evidence grounding is essential for reliable multimodal reasoning [9, 10].

Although recent studies have advanced fine-grained evidence grounding [11, 12], two limitations remain. First, existing methods often yield fragmented evidence grounding: evidence can be grounded as separate documents [13], entities [14], regions [15], pixels [15], or actions [16], but explicit evidence-anchor correspondences between heterogeneous evidence units and evidence anchors are rarely modeled. As a result, the grounding result remains weakly structured and less interpretable, especially when the evidence contains complementary, redundant, or noisy signals. Second, grounded modalities remain vulnerable to modality dominance. Stronger or denser modalities may dominate the fused representation, while weaker but complementary modalities can be under-utilized [17, 18]. Existing fusion methods mitigate this issue with adaptive weights [19], ensemble decisions [20], modality experts [21, 22], or complementarity modeling [18], but they mainly rely on direct weighting or decision aggregation and do not explicitly estimate each modality's utility gain over its own baseline state. These limitations motivate the following question:

How can heterogeneous multimodal evidence be grounded into explicit evidence-anchor correspondences without allowing dominant modalities to suppress complementary evidence?

To address this question, we propose NuGAT, a general evidence-grounded multimodal alignment and fusion framework with Attention Transport (AT) and Nash Utility Fusion (NUF). AT formulates evidence grounding as structured evidence-anchor correspondence learning. It treats modality-specific evidence units as transferable sources and evidence anchors as grounding targets, learning a transport plan that specifies how much information each anchor receives from each evidence unit. Thus, the grounding result is not a flat evidence set but an explicit cor-

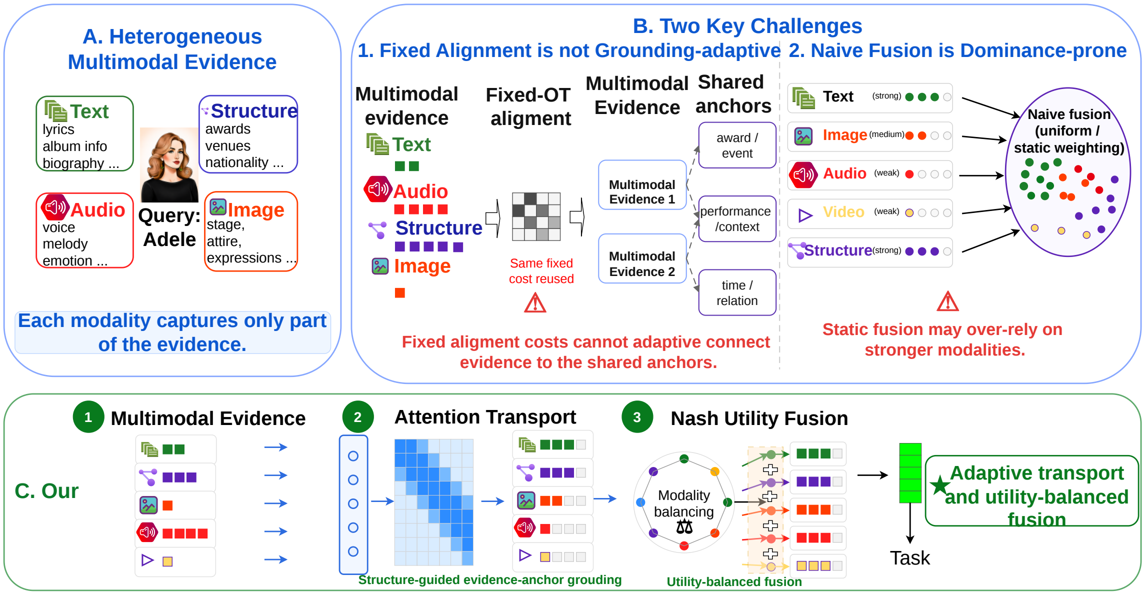


Figure 1: Motivation for NuGAT. AT grounds heterogeneous multimodal evidence as explicit evidence-anchor correspondences, while NUF mitigates modality dominance through utility-controlled subspace retention.

respondence structure traceable to the original multimodal evidence. NUF addresses modality dominance after grounding. Inspired by Nash bargaining [23], which balances utility gains among multiple players and has been applied to multi-task and auxiliary learning [24, 25], NuGAT treats each modality as a utility-bearing modality. Its utility measures the agreement between the grounded modality representation and the fused evidence representation, while its disagreement state denotes the modality-specific baseline before cooperative fusion. Rather than using utility gain as a scalar fusion weight, NUF uses it to control modality-specific subspace retention. This preserves informative directions from reliable modalities and suppresses noisy or weakly grounded directions. The main contributions are summarized as follows[2]:

- We propose NuGAT, a general evidence-grounded multimodal alignment and fusion framework for addressing fragmented evidence grounding and modality dominance.
- We introduce Attention Transport to ground modality-specific evidence units into evidence anchors, producing explicit and interpretable evidence-anchor correspondences.
- We design Nash Utility Fusion to mitigate modality dominance by using utility gain to control subspace retention rather than direct scalar weighting.

[2]Source codes are available on `https://github.com/howieWu9/NuGAT`.

- We conduct experiments on multimodal QA and concept verification, demonstrating consistent improvements in evidence-grounded reasoning and verification.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 presents NuGAT. Section 4 reports the experimental analyses. Section 5 concludes the paper.

2. Related work

2.1. Multimodal Evidence Grounding for Reasoning

Multimodal reasoning requires models to ground query-relevant multimodal evidence from heterogeneous modalities, including text, images, audio, video, and structural, so that the reasoning process can be supported by grounded multimodal evidence [1, 2]. Early MLLMs mainly perform coarse-grained multimodal understanding [3, 4], which limits their ability to ground cross-modal correspondences and multi-step evidence dependencies [5, 6, 7, 8]. Therefore, as reasoning tasks become more complex, multimodal evidence grounding must move beyond coarse context-level understanding toward fine-grained grounding [9, 10].

Recent studies have explored increasingly fine-grained evidence grounding for multimodal reasoning [11, 12, 26, 27, 28]. Existing methods ground evidence at different granularities, including document-level grounding [13], entity and passage-level grounding [14], pixel-level response grounding [29], region-level spatial grounding [15], and action-level grounding [16]. These studies show that grounding concrete evidence units is important for connecting reasoning with multimodal evidence. Meanwhile, cross-modal matching [30] and cross-modal alignment methods further improve evidence grounding across heterogeneous modalities through robust hashing [31], contrastive alignment [32], semantic distillation [33], or relation-aware evidence modelingg [34]. However, existing methods still mainly yield fragmented evidence grounding. They can ground documents, entities, regions, pixels, or actions, but the grounded evidence is often represented as separate evidence units rather than explicit correspondences between heterogeneous evidence units and shared evidence anchors. Moreover, many methods rely on predefined associations, making it difficult to obtain structured and interpretable multimodal grounding when the evidence contains complementary, redundant, or noisy signals.

To address this limitation, NuGAT formulates evidence grounding as structured evidence-anchor correspondence learning. Attention Transport treats modality-specific evidence units as transferable sources and evidence anchors as grounding targets. It learns a transport plan that specifies how much information each anchor receives from each evidence unit, thereby grounding heterogeneous evidence into explicit evidence-anchor correspondences. This structured transport process improves interpretability because each learned correspondence directly links an evidence anchor to its contributing multimodal evidence units.

2.2. Nash Bargaining for Multimodal Fusion

After evidence grounding establishes explicit evidence-anchor correspondences, multimodal fusion must integrate grounded evidence without allowing unreliable or dominant modalities to overwhelm complementary evidence. Although structured grounding makes evidence-anchor correspondences explicit, modalities may still differ in reliability, informativeness, and noise level [18]. A modality with stronger features or denser evidence may dominate the fused representation, even when other modalities provide complementary evidence [17]. Balanced multimodal learning has therefore been widely studied to mitigate modality dominance in multimodal fusion [19]. Existing multimodal fusion methods coordinate heterogeneous modalities through relation-guided adaptive fusion [20], modality-split ensemble inference [21], modality knowledge experts [22], or complementarity-guided fusion [18]. These methods improve modality coordination by estimating fusion weights, ensemble decisions, expert outputs, or complementarity scores. However, they still mainly rely on direct weighting or decision aggregation and do not explicitly estimate each modality's utility gain over its own baseline state after grounding. Consequently, dominant modalities may still absorb excessive information in the fused evidence representation.

Nash bargaining provides a cooperative principle for balancing utility gains among multiple players [23]. Given K players with utility u_i and disagreement utility d_i, the Nash bargaining solution maximizes the product of utility gains, which is commonly written in log form as

$$\max \sum_{i=1}^{K} \log(u_i - d_i), \quad \text{s.t.}; u_i > d_i, \ \forall i. \tag{1}$$

This formulation discourages solutions in which one player obtains a large improvement while others receive limited gains. In machine learning, Nash-style optimization has been used for multi-task gradient combination [24] and auxiliary learning [25]. However, these methods usually define players as losses, rather than modality evidence sources in multimodal fusion.

NuGAT extends this principle to grounded multimodal fusion. Each modality is treated as a utility-bearing modality, whose utility is defined by the alignment quality between its grounded and the fused evidence representation. The disagreement utility represents the modality-specific baseline before cooperative fusion. Unlike direct scalar weighting, the resulting utility gain controls modality-specific subspace retention before fusion. This design preserves informative subspace directions from reliable modalities while suppressing noisy or weakly grounded directions. It also improves interpretability, because modality utilities and retained subspace directions reveal how each modality contributes to the final grounded evidence representation.

3. Method

3.1. Problem Formulation

In this paper, we mainly focus on the evidence grounding and multimodal fusion stage of multimodal reasoning. Let $\mathcal{M}$ denote the modality set. For each modality $m \in \mathcal{M}$, the input evidence representation is $X_m \in \mathbb{R}^{n_m \times d_m}$, where n_m is the number of evidence units and d_m is the original feature dimension. Our goal is to construct a shared multimodal evidence representation $Y \in \mathbb{R}^{n \times d}$, where n is the number of shared evidence anchors and d is the unified hidden dimension. Different modalities may contain different numbers of evidence units and lie in different feature spaces, so each modality-specific representation X_m is first aligned to the shared anchor space, producing Y_m, then fused to obtain the final shared representation Y. which serves as the multimodal evidence representation for downstream tasks.

3.2. Overview

NuGAT follows two principles. First, evidence grounding should not return a flat evidence set, but should establish explicit evidence-anchor correspondences, where each evidence anchor selectively receives information from informative modality-specific evidence units. Second, multimodal fusion should mitigate modality dominance by retaining reliable modality information and suppressing noisy or weakly grounded signals.

The first module, Attention Transport, learns a transport plan that specifies how much information each evidence anchor receives from each modality-specific evidence unit. Thus, the transport plan yields structured evidence-anchor correspondences rather than simple evidence

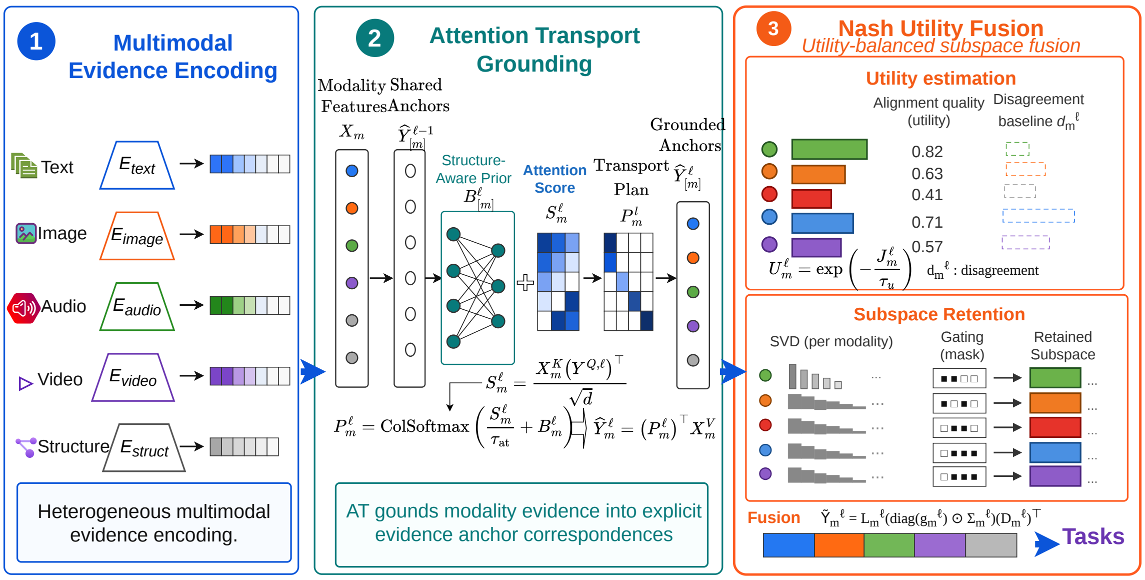


Figure 2: Overall framework of NuGAT. AT produces explicit evidence-anchor correspondences through attention-guided and structure-aware transport, while NUF mitigates modality dominance via utility-guided subspace retention and fuses grounded modalities for downstream tasks.

selection. To incorporate prior correspondence cues, we introduce a soft compatibility prior based on entity correspondence and pre-computed structural similarity.

The second module, Nash Utility Fusion, evaluates the reliability of each grounded modality representation and converts it into a modality utility. This utility is compared with a modality-specific disagreement state to obtain a utility gain. Instead of using the utility gain as a scalar fusion weight, NuGAT uses it to control modality subspace retention before fusion.

At ℓ, each modality X_m is first transported to the previous shared state $Y^{\ell-1}$, producing an aligned representation $\widehat{Y}^{\ell}_{m}$. Then, Nash utility controls the retained principal subspace of $\widehat{Y}^{\ell}_{m}$, and the filtered modality representations are fused into the next shared state Y^{ℓ}.

3.3. Attention-based Transport

Classical Optimal Transport (OT) seeks a transport plan P that transfers mass from a source distribution to a target distribution with minimum cost. Given a cost matrix $C \in \mathbb{R}^{n_m \times n}$, the entropy-regularized balanced OT problem is commonly written as

$$P^{\star} = \arg \min_{P \in \mathcal{U}_{\text{bal}}} \left(\langle P, C \rangle + \tau_{\text{ot}} \text{KL}(P) \right), \tag{2}$$

where the feasible set constrains both source-side and target-side marginals:

$$\mathcal{U}_{\text{bal}} = \left\{ P \in \mathbb{R}_+^{n_m \times n} : P\mathbf{1}_n = \mathbf{1}_{n_m},\ P^\top \mathbf{1}_{n_m} = \mathbf{1}_n \right\}, \tag{3}$$

The first constraint forces every source unit to be transported, while the second constraint requires every target unit to receive a fixed amount of mass.

This balanced formulation is not fully suitable for multimodal evidence grounding. For a given query, not every modality-specific evidence unit is useful. Some evidence units can be irrelevant, redundant, or noisy. If all source units are forced to contribute, the shared representation may absorb unnecessary information. Therefore, instead of using balanced OT as our grounding operator, we design an AT mechanism that only constrains the information received by each shared anchor.

For modality m at ℓ, the attention transport plan is denoted as $P_m^\ell \in \mathbb{R}_+^{n_m \times n}$, where $P_{m,ij}^\ell$ denotes how much information the j-th shared anchor receives from the i-th evidence unit of modality m. Our one-sided feasible set is

$$\mathcal{U}_{\text{at}} = \left\{ P \in \mathbb{R}_+^{n_m \times n} : P^\top \mathbf{1}_{n_m} = \mathbf{1}_n \right\}, \tag{4}$$

This constraint normalizes the received information for each shared anchor, but it does not force every source evidence unit to be used. Thus, each anchor can selectively gather informative evidence from modality m. To make this selection query-adaptive, we use attention to construct the transport reward. The query, key, and value representations are computed as

$$Y^{Q,\ell} = Y^{\ell-1} W_Y^Q, \qquad X_m^K = X_m W_m^K, \qquad X_m^V = X_m W_m^V, \tag{5}$$

where

$$W_Y^Q \in \mathbb{R}^{d \times d}, \qquad W_m^K \in \mathbb{R}^{d_m \times d}, \qquad W_m^V \in \mathbb{R}^{d_m \times d}, \tag{6}$$

Here, $Y^{Q,\ell}$ describes what information each shared anchor intends to retrieve, X_m^K describes what semantic information each modality sample can provide, and X_m^V is the content that is actually transferred after dimension alignment. Therefore, keys are used for matching, while values are used for information transfer. The attention reward matrix is

$$S_m^\ell = \frac{X_m^K \left(Y^{Q,\ell}\right)^\top}{\sqrt{d}}, \qquad S_m^\ell \in \mathbb{R}^{n_m \times n}, \tag{7}$$

The entry $S_{m,ij}^\ell$ measures the semantic matching degree between the i-th evidence unit of modality m and the j-th shared anchor.

However, attention scores capture only current feature-level similarity. In evidence grounding, entity correspondence and high-level structural compatibility provide useful prior cues for transporting modality-specific evidence units to evidence anchors. We therefore introduce a soft compatibility prior $\exp\bigl(B_m^\ell\bigr)$, where

$$B_{m,ij}^\ell = \log \frac{\exp\left(\mathrm{sim}_{\mathcal{G}_q}(x_{m,i}, y_j)\right)}{\sum_{a=1}^{n_m} \exp\left(\mathrm{sim}_{\mathcal{G}_q}(x_{m,a}, y_j)\right)}, \tag{8}$$

The structural similarity is defined as

$$\mathrm{sim}_{\mathcal{G}_q}(x_{m,i}, y_j) = \mathbb{I}(e(x_{m,i}) = e(y_j)) + s_q^{\mathrm{LLM}}(x_{m,i}, y_j), \tag{9}$$

where $\mathcal{G}_q$ denotes the query-related evidence context, $e(\cdot)$ returns the entity associated with an evidence unit or a shared anchor, and $s_q^{\mathrm{LLM}}(x_{m,i}, y_j)$ denotes the query-conditioned structural compatibility between the evidence unit and the shared support anchor, estimated offline by a MMLM before NuGAT transport. The first term captures entity correspondence, while the second term provides a soft compatibility prior beyond exact entity matching.

The prior B_m^ℓ does not hard-code the transport plan. When reliable structural evidence is available, it serves as a relation-guided soft bias. When structural evidence is missing or unreliable, it approaches a column-uniform prior and has limited effect after column-wise normalization. In this case, the transport plan is mainly determined by the attention reward S_m^ℓ.

Given the attention reward and the soft compatibility prior, we define the AT plan by

$$P_m^\ell = \arg\max_{P \in \mathcal{U}_{\mathrm{at}}} \left(\langle P, S_m^\ell \rangle - \tau_{\mathrm{at}} \mathrm{KL}\bigl(P \,\|\, \exp\bigl(B_m^\ell\bigr)\bigr)\right), \tag{10}$$

The first term encourages each evidence anchor to collect evidence units with high semantic matching scores. The second term regularizes the transport plan toward the soft compatibility prior. Thus, the learned transport plan captures and structure prior preference and attention.

The above one-sided problem has a closed-form solution:

$$P^{\ell}_{m,ij} = \frac{\exp\left(S^{\ell}_{m,ij}/\tau_{\mathrm{at}} + B^{\ell}_{m,ij}\right)}{\sum_{a=1}^{n_m}\exp\left(S^{\ell}_{m,aj}/\tau_{\mathrm{at}} + B^{\ell}_{m,aj}\right)}, \tag{11}$$

Equivalently,

$$P^{\ell}_{m} = \mathrm{ColSoftmax}\left(\frac{S^{\ell}_{m}}{\tau_{\mathrm{at}}} + B^{\ell}_{m}\right), \qquad P^{\ell}_{m} \in \mathbb{R}^{n_m \times n}, \tag{12}$$

where $\mathrm{ColSoftmax}(\cdot)$ denotes column-wise softmax normalization. Each column of P^{ℓ}_{m} is therefore a normalized AT distribution for one shared anchor. The learned transport plan P^{ℓ}_{m} also provides an interpretable grounding trace. Specifically, $P^{\ell}_{m,ij}$ indicates how much the j-th shared evidence anchor relies on the i-th evidence unit of modality m. Therefore, NuGAT can trace each grounded evidence anchor back to its contributing multimodal evidence units.

Finally, modality m is aligned to the shared anchor space by transporting its value:

$$\widehat{Y}^{\ell}_{m} = \left(P^{\ell}_{m}\right)^{\top} X^{V}_{m}, \qquad \widehat{Y}^{\ell}_{m} \in \mathbb{R}^{n \times d}, \tag{13}$$

Thus, $\widehat{Y}^{\ell}_{m}$ is the modality anchor representation produced by attention-based transport.

3.4. Nash Utility Fusion

After attention-based transport, all modalities are represented in the same evidence-anchor space. The key fusion problem is determining how much information each grounded modality should retain. Simple scalar-weighted summation may lead to modality dominance in the fused state. To mitigate this, we use Nash utility gain to control modality-specific subspace retention, rather than using modality utilities as direct scalar fusion weights.

The modality utilities are initialized uniformly as $U^{0}_{m} = 1.$, indicating that all modalities are treated equally before any alignment feedback is observed. At ℓ, the retention ratio of modality m is computed from the previous utilities:

$$\alpha^{\ell}_{m} = \frac{U^{\ell-1}_{m}}{\sum_{r\in\mathcal{M}} U^{\ell-1}_{r}}, \tag{14}$$

Here, α^{ℓ}_{m} indicates the information retention strength of modality m at the current. Using the previous utility avoids a circular dependency between the current fused state and the current utility evaluation.

For each aligned representation $\widehat{Y}_m^\ell$, we first perform Singular Value Decomposition (SVD):

$$\widehat{Y}_m^\ell = L_m^\ell \Sigma_m^\ell (D_m^\ell)^\top, \qquad r_{\max} = \min(n, d). \tag{15}$$

Here, L_m^ℓ and D_m^ℓ denote the left and right singular directions, respectively, and Σ_m^ℓ is the diagonal singular-value matrix. Since the magnitude ordering of singular values does not necessarily reflect their contribution to the fused evidence state, fixed top-k truncation may preserve redundant or noisy directions while removing useful but lower-magnitude ones [35]. We therefore learn a subspace mask over singular directions, enabling direction-level retention to be adapted to modality reliability.

$$z_m^\ell \sim \mathcal{N}\left(\alpha_m^\ell \mathbf{1}_{r_{\max}} - \beta_m^\ell, \tau_g^2 I_{r_{\max}}\right). \tag{16}$$

Here, $z_m^\ell \in \mathbb{R}^{r_{\max}}$ is the continuous latent variable for mask generation, $\beta_m^\ell \in [0,1]^{r_{\max}}$ is a learnable direction-wise retention threshold, $I_{r_{\max}}$ is the identity matrix, and τ_g controls the sampling variance. The mean $\alpha_m^\ell \mathbf{1}_{r_{\max}} - \beta_m^\ell$ defines the retention tendency of each singular direction, where α_m^ℓ reflects the modality-level retention strength. The latent variable is mapped into the subspace mask:

$$g_m^\ell = \mathbb{I}\left(z_m^\ell \geq 0\right), \qquad g_m^\ell \in \{0,1\}^{r_{\max}}. \tag{17}$$

For the i-th singular direction, $g_{m,i}^\ell = 1$ retains this direction, while $g_{m,i}^\ell = 0$ suppresses it. The mask is applied to the singular-value subspace:

$$\widetilde{Y}_m^\ell = L_m^\ell \left(\operatorname{diag}\left(g_m^\ell\right) \odot \Sigma_m^\ell\right) (D_m^\ell)^\top. \tag{18}$$

The operator $\odot$ denotes the Hadamard product. Since Σ_m^ℓ is diagonal, $\operatorname{diag}(g_m^\ell) \odot \Sigma_m^\ell$ sets suppressed singular directions to zero. In this way, α_m^ℓ controls how much subspace information modality m tends to preserve, while g_m^ℓ determines which singular directions are retained for reconstruction.

After obtaining Y^ℓ, we evaluate the alignment quality of each modality. The content transported from modality m to the i-th anchor should be closer to the corresponding fused anchor than to other anchors. We define the anchor-level contrastive alignment loss as J_m^ℓ. A smaller

J_m^ℓ means that modality m is better aligned with the current shared state. The alignment loss is converted into a positive utility U_m^ℓ:

$$U_m^\ell = \exp\left(-\frac{J_m^\ell}{\tau_u}\right), \qquad J_m^\ell = -\frac{1}{m}\sum_{i=1}^{m}\log\frac{\exp\left(\mathrm{sim}\left(\hat{Y}_m^\ell, Y^\ell\right)\right)}{\sum_{j=1}^{m}\exp\left(\mathrm{sim}\left(\hat{Y}_m^\ell, Y^\ell\right)\right)}, \tag{19}$$

Thus, a well-aligned modality receives a larger utility and will retain more principal subspace information in the next. A poorly aligned modality receives a smaller utility and will be more strongly suppressed.

Finally, we maximize the Nash product of modality utilities across layers, which is equivalent to minimizing

$$\mathcal{L}_{\mathrm{nash}} = -\sum_{\ell=1}^{L}\sum_{m\in\mathcal{M}}\log\left(U_m^\ell + \epsilon\right). \tag{20}$$

This objective encourages all modalities to contribute useful evidence to the shared state, while the utility-controlled subspace gates prevent noisy modalities from directly dominating the fusion process.

4. Experiments

We evaluate NuGAT as a multimodal evidence grounding and fusion module for evidence-grounded reasoning. Given a reasoning instance and its available multimodal evidence context, NuGAT grounds heterogeneous textual, audio, visual, and structural evidence into explicit evidence-anchor correspondences, and produces a fused grounded evidence representation for downstream task. We evaluate NuGAT from two complementary perspectives. Multimodal QA examines whether the grounded evidence representation improves open-ended reasoning across audio, video, and audio-visual questions. Multimodal concept verification further examines whether NuGAT can distinguish evidence-supported candidates from noisy ones. These two tasks respectively assess the downstream usefulness of grounded evidence and the intrinsic ability of NuGAT to verify cross-modal evidence consistency.

4.1. Datasets

For multimodal QA, we follow the previous evaluation setting [28] and use AudioCaps-QA [36], VCGPT [37], and VALOR [38], which focus on audio QA, video understanding, and audio-visual reasoning, respectively. AudioCaps-QA [36] contains 313 audio QA samples adapted from AudioCaps, covering 0.86 hours of audio with an average duration of 9.86 seconds; it is used to evaluate whether NuGAT can construct audio-grounded evidence. VCGPT [37] is built on ActivityNet-200 [39] and contains 500 curated test samples for detailed video understanding, serving as a benchmark for video-grounded evidence construction. For VALOR [38], we adopt the commonly used VALOR-32K setting, which contains 32K synchronized audio-visual clips with human-authored captions and evaluates whether NuGAT can fuse complementary audio-visual evidence. For multimodal concept verification, we use SciMKG [40], a multimodal educational knowledge graph for Chinese K-12 science education covering biology, physics, and chemistry. It contains 1,356 knowledge points, 34,630 multimodal concepts, 403,400 triples, 10,527 images, 10,425 videos, and 34,630 audio files.

4.2. Baselines and Evaluation Metrics

Multimodal QA. For multimodal QA, we follow the evaluation protocol of M^3KG-RAG [28], where the answerer is fixed as VideoLLaMA2 or Qwen2.5-Omni and each method only changes the external evidence provided to the answerer. Therefore, performance differences mainly reflect the quality of evidence grounding and fusion rather than the generation ability of the answerer. We compare NuGAT with Wikidata [41], VTKG [42], M^2ConceptBase [26], VAT-KG [27], and M^3KG-RAG [28]. Following the same protocol, we report Model-as-Judge (M.J.) scores, where a Llama-3 judge scores each generated answer conditioned on the query and reference answer. M.J. evaluates whether the generated answer is correct, relevant, and supported by the multimodal evidence. We repeat the evaluation over multiple runs and report standard deviations to reflect result stability.

Multimodal concept verification. For concept extraction and verification, this paper follows the evaluation protocol of prior work and compares against representative extraction and NER baselines [40], including Decomposed-QA [43], GPT-NER [44], LinkNER [45], UniversalNER [46], Self-Improving [47], and the original multimodal educational pipeline [40]. Candidate generation is kept unchanged, and NuGAT is evaluated as a multimodal verifier and ranker using surrounding text, image, video, audio, and relational evidence. Following the original protocol [40], we report Precision, Recall, F1-score, Accuracy (ACC), Matthews Correlation Coefficient (MCC), and Area Under the ROC Curve (AUC).

4.3. Implementation Details

All experiments are implemented in PyTorch within a Python Conda environment and conducted on an H100 computing cluster. The computing node runs Ubuntu Server 22.04 LTS and is equipped with 72 CPU cores, 463GB RAM, and two NVIDIA H100 GPUs, each with 80GB memory. CUDA 12.2 and Anaconda 3 are used to configure the experimental environment. Specifically, we use the Adam optimizer [48] with a learning rate of 1×10^{-4}, a batch size of 1024, and a maximum of 1000 epochs. Entity embeddings, relation embeddings, and trainable modality-specific modules are initialized with Xavier uniform initialization. Unless otherwise specified, the hidden dimension of the scoring module is fixed to 256.

For modality encoding, we use the released modality features when available to ensure fair comparison [20, 18, 12]. For each modality m, $\mathrm{Enc}_m(\cdot; \Theta_m)$ produces the modality-specific representation $\mathbf{X}_m$. NuGAT preserves its original feature dimension d_m before alignment, rather than projecting all modalities into a predefined shared space. The main hyperparameters are selected on the validation set and fixed for testing.

4.4. Main Results

4.4.1. Evidence-Grounded Multimodal Question Answering

The first experiment evaluates whether the grounded multimodal evidence produced by NuGAT improves open-ended QA. The QA benchmarks provide questions, multimodal inputs, and reference answers. Under the fixed-answerer setting, each method provides evidence representations from the available multimodal evidence context, while the answer generator remains unchanged. Therefore, any improvement mainly comes from stronger evidence grounding, alignment, and fusion rather than from a stronger answer generator.

Table 1 shows that NuGAT consistently improves multimodal QA under both fixed answerers and all evidence settings. Since the answerer is unchanged, these gains cannot be attributed to stronger generation ability, but instead indicate that NuGAT provides more effective grounded evidence for the same MLLM. Compared with the strongest baseline, M^{3}KG-RAG, NuGAT achieves average improvements of 3.38 and 3.35 M.J. points with VideoLLaMA2 and Qwen2.5-Omni, respectively. This consistent improvement across two different answerers suggests that the benefit of NuGAT is not model-specific, but comes from better evidence grounding and fusion. The gains on AudioCaps-QA and VALOR are particularly informative.

Table 1: Multimodal QA performance measured by Model-as-Judge (M.J.; higher is better). The best and second-best results are shown in **bold** and underlined, respectively. Gray rows report NuGAT and the five-run deviation.

MLLM	Methods	Audio QA	Video QA	Audio-Visual QA
		AudioCaps-QA	VCGPT	VALOR
VideoLLaMA2 [49]	None	43.13	39.09	25.66
	Wikidata [41]	43.58	38.58	26.43
	VTKG [42]	43.02	38.88	25.92
	M^{2}ConceptBase [26]	42.19	39.31	25.93
	VAT-KG [27]	44.60	39.42	28.30
	M^{3}KG-RAG [28]	<u>53.23</u>	<u>39.92</u>	<u>29.25</u>
	OUR	**57.12**	**43.26**	**32.15**
	Std.	±1.24	±1.19	±1.08
Qwen2.5-Omni [5]	None	49.00	42.21	32.42
	Wikidata [41]	49.78	40.82	30.28
	VTKG [42]	48.95	42.96	32.70
	M^{2}ConceptBase [26]	49.78	42.78	32.31
	VAT-KG [27]	51.30	43.50	35.44
	M^{3}KG-RAG [28]	<u>60.77</u>	<u>44.35</u>	<u>44.67</u>
	OUR	**63.88**	**47.94**	**48.03**
	Std.	±1.54	±1.23	±1.15

AudioCaps-QA relies on sound-related evidence, while VALOR requires coordinated audio-visual grounding. The improvements on these two benchmarks suggest that AT can better connect query-relevant evidence units to evidence anchors, rather than treating multimodal evidence as a flat context. Meanwhile, the stable gains across audio, video, and audio-visual settings indicate that NUF helps reduce modality dominance by retaining reliable modality-specific subspaces and suppressing noisy or weakly grounded directions. Therefore, the QA results support our central claim: structured evidence-anchor grounding and utility-guided subspace retention produce more useful grounded evidence representations for multimodal QA.

4.4.2. Multimodal Concept Verification

The second experiment evaluates whether NuGAT can support evidence-based concept verification. Unlike multimodal QA, which measures downstream answer generation, this task directly examines whether candidate concepts extracted from multimodal educational resources are supported by their surrounding evidence. Following the benchmark protocol [40], we keep the candidate generation stage unchanged and use NuGAT only as a multimodal verifier and ranker. Given a candidate concept and its surrounding textual, visual, video, audio, and relational evidence, NuGAT estimates its cross-modal evidence support and ranks reliable candidates above noisy ones.

Table 2: Concept verification performance on the multimodal educational benchmark [40] (higher is better). The best and second-best results are shown in **bold** and underlined, respectively. Gray rows report NuGAT and five-run standard deviations.

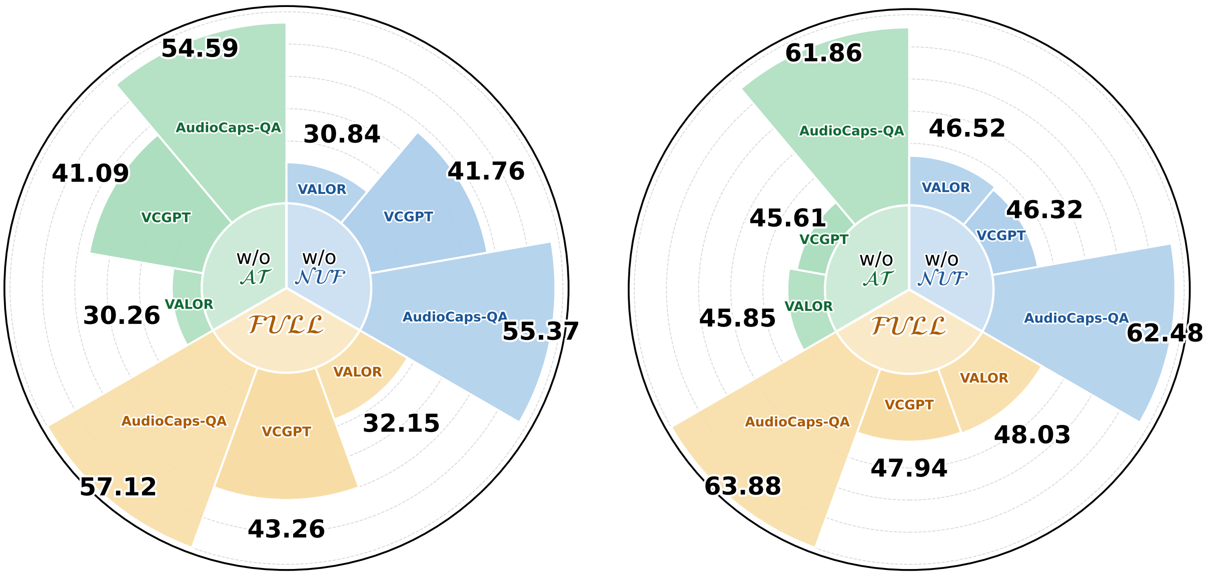
Method	Precision	Recall	F1-score	ACC	MCC	AUC
Decomposed-QA [43]	0.935	0.570	0.712	0.977	0.741	0.885
GPT-NER [44]	0.530	**0.931**	0.681	0.923	0.621	0.938
LinkNER [45]	0.855	0.644	0.734	0.981	0.732	0.912
UniversalNER [46]	0.873	0.617	0.718	0.974	0.722	0.929
Self-Improving [47]	0.911	0.532	0.670	0.972	0.725	0.935
SciMKG [40]	0.947	0.702	0.803	0.986	0.729	0.985
NuGAT	**0.958**	0.781	**0.860**	**0.989**	**0.860**	**0.991**
Std.	±0.006	±0.012	±0.010	±0.002	±0.013	±0.002

Table 2 shows that NuGAT provides a stronger precision–recall trade-off for concept verification. GPT-NER achieves the highest recall, but its much lower precision suggests that it retains many weakly supported candidates. In contrast, NuGAT improves recall over the original SciMKG pipeline while achieving the highest precision, indicating that it increases coverage without sacrificing evidence reliability. This leads to a clear F1-score improvement from 0.803 to 0.860, showing that structured evidence grounding and reliability-aware fusion help separate valid concepts from noisy candidates. The improvement in MCC is particularly important. Since MCC is more robust to class imbalance than accuracy, the gain from 0.729 to 0.860 suggests that NuGAT improves both positive and negative verification decisions, rather than only benefiting from the dominant class. The highest AUC further indicates that the learned verification score provides a more reliable ranking of candidate concepts across decision thresholds. These results show that NuGAT is effective not only as a downstream evidence enhancer but also as a multimodal evidence verifier: AT aligns local multimodal evidence with concept-level structural semantics, while NUF reduces the influence of weakly supported modalities.

4.5. Ablation Analysis

To isolate the contribution of each component, we conduct ablation and diagnostic analyses on AudioCaps-QA, VCGPT, and VALOR using the same fixed-answerer QA protocol as the main experiment. The answerer is kept unchanged, and only the evidence grounding or fusion strategy is modified. Thus, performance differences directly reflect the design in NuGAT.

4.5.1. Model Breakdown



(a) Model breakdown with VideoLLaMA2.

(b) Model breakdown with Qwen2.5-Omni.

Figure 3: Model breakdown of NuGAT on multimodal QA. Each panel reports Model-as-Judge (M.J.) scores on AudioCaps-QA, VCGPT, and VALOR under a fixed MLLM answerer.

We conduct a component breakdown to examine the roles of $\mathcal{AT}$ and $\mathcal{NUF}$ under the fixed-answerer QA setting. The w/o $\mathcal{AT}$ variant removes attention-based transport and directly aggregates modality-specific evidence without learning evidence-anchor correspondences, while the w/o $\mathcal{NUF}$ variant replaces Nash Utility Fusion with uniform aggregation after evidence grounding. As shown in Fig. 3a and Fig. 3b, both variants consistently underperform the full model on AudioCaps-QA, VCGPT, and VALOR, confirming that structured grounding and utility-guided fusion are complementary. The larger drop of w/o $\mathcal{AT}$ indicates that direct aggregation cannot sufficiently align heterogeneous textual, audio, visual, and structural evidence into a coherent grounded representation. This supports the necessity of explicit evidence-anchor correspondences before fusion. The degradation of w/o $\mathcal{NUF}$ further shows that even after evidence grounding, uniform aggregation is insufficient because noisy or weakly grounded modalities may still dominate the fused evidence representation. Therefore, $\mathcal{AT}$ mainly improves the structure and interpretability of evidence grounding, whereas $\mathcal{NUF}$ mitigates modality dominance through utility-guided subspace retention, leading to more reliable evidence for MLLM-based answer generation.

4.5.2. Ablation of Attention Transport

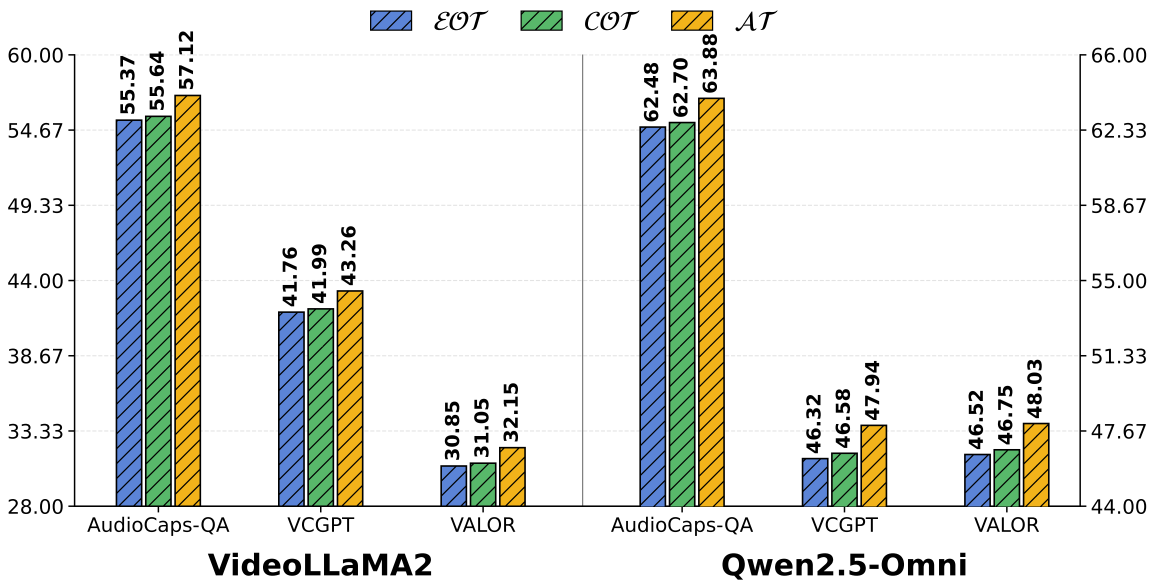


Figure 4: Transport ablation. $\mathcal{EOT}$ and $\mathcal{COT}$ use fixed squared-Euclidean and cosine distances, respectively, whereas NuGAT learns an attention-based evidence-anchor matching for transport.

To isolate the effect of transport scoring, we replace the learnable evidence-anchor matching score in $\mathcal{AT}$ with two fixed-distance OT variants: $\mathcal{EOT}$ with squared Euclidean distance $c(x_m, y) = \|\rho_m(x_m) - \rho_y(y)\|_2^2$, and $\mathcal{COT}$ with cosine distance $c(x_m, y) = 1 - \cos(\rho_m(x_m), \rho_y(y))$. As shown in Fig. 4, $\mathcal{EOT}$ and $\mathcal{COT}$ perform similarly, indicating that simply changing the predefined geometric metric brings limited benefit. In contrast, NuGAT achieves the best results by learning an attention-based evidence-anchor matching score. Since all variants share the same transport framework and differ mainly in how evidence-anchor matching is computed, the improvement can be attributed to task-relevant transport scoring rather than fixed geometric distances. Beyond performance gains, the learned transport plan also improves interpretability. Each entry indicates how much information an evidence anchor receives from a modality-specific evidence unit, thereby providing an explicit evidence-anchor correspondence for tracing the grounding process.

4.5.3. Ablation of Nash Utility Fusion

We further examine whether multimodal fusion requires adaptive reliability modeling after evidence grounding. $\mathcal{UWF}$ uniformly aggregates grounded modalities, whereas $\mathcal{SWF}$ uses

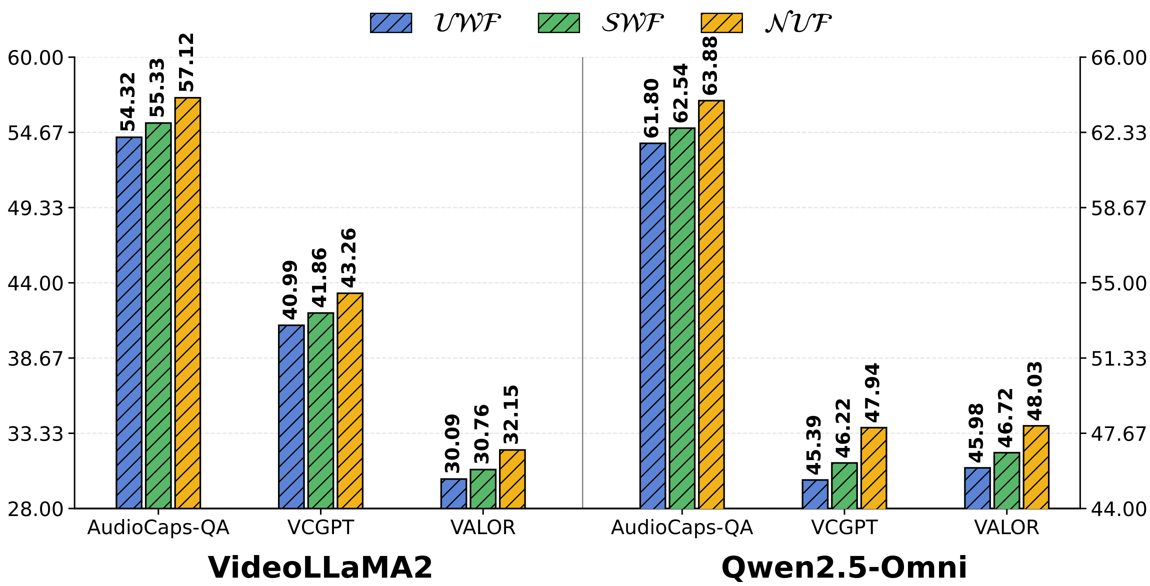


Figure 5: Ablation of $\mathcal{NUF}$. Compared with uniform weighting ($\mathcal{UWF}$) and static weighting ($\mathcal{SWF}$).

globally fixed modality weights. As shown in Fig. 5, $\mathcal{SWF}$ consistently outperforms $\mathcal{UWF}$ on all three QA benchmarks, indicating that modality importance should not be treated as uniform. However, $\mathcal{NUF}$ achieves the best M.J. scores under both VideoLLaMA2 and Qwen2.5-Omni, improving over $\mathcal{SWF}$ by an average of 1.53 and 1.46, respectively. These results reveal a stronger conclusion: although global weights capture coarse modality importance, they are still insufficient for evidence-grounded QA, where modality reliability changes across questions and evidence contexts. For example, audio evidence is crucial for AudioCaps-QA, while VALOR requires coordinated audio-visual evidence. A fixed weighting scheme cannot adapt to such instance-level reliability shifts and may still suffer from modality dominance. Since all variants use the same attention-grounded representations and differ only in the fusion strategy, the gains of $\mathcal{NUF}$ can be attributed to utility-guided subspace retention, which preserves reliable modality directions while suppressing noisy or weakly grounded ones.

4.5.4. Modality Contribution Ablation

Table 3 analyzes how different evidence modalities contribute to evidence-grounded QA, where S, T, A, and V denote structural, textual, audio, and visual/video evidence, respectively. The full model achieves the best M.J. scores across all benchmarks and both fixed answerers, indicating that NuGAT benefits from complementary multimodal evidence rather than relying on a single dominant modality.

Table 3: Modality contribution analysis on evidence-grounded QA. S/T/A/V denote structure, text, audio, and video evidence. The metric is Model-as-Judge (M.J.), and higher scores are better.

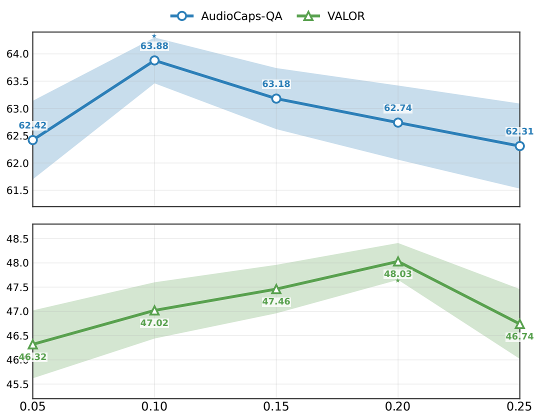
Benchmark	Variant	S	T	A	V	VideoLLaMA2	Qwen2.5-Omni
AudioCaps-QA	T only		✓			49.18	56.02
	A only			✓		52.64	59.37
	w/o S		✓	✓		54.21	61.38
	w/o T	✓		✓		55.03	62.08
	w/o A	✓	✓			51.46	58.24
	Full	✓	✓	✓		**57.12**	**63.88**
VCGPT	T only		✓			39.55	43.38
	V only				✓	41.06	45.44
	w/o S		✓		✓	42.22	46.80
	w/o T	✓			✓	41.70	46.12
	w/o V	✓	✓			40.43	44.58
	Full	✓	✓		✓	**43.26**	**47.94**
VALOR	T only		✓			27.82	41.46
	A only			✓		28.66	42.72
	V only				✓	28.10	42.05
	A+V			✓	✓	30.42	45.67
	w/o S		✓	✓	✓	31.28	46.74
	w/o T	✓		✓	✓	30.76	46.18
	w/o A	✓	✓		✓	29.72	45.48
	w/o V	✓	✓	✓		30.08	45.86
	Full	✓	✓	✓	✓	**32.15**	**48.03**

The results also reveal that modality importance is task-dependent. AudioCaps-QA relies more heavily on audio evidence: the audio-only variant outperforms the text-only variant, and removing audio causes the largest performance drop. In contrast, VCGPT depends more on visual/video evidence, as the visual-only variant is stronger than the text-only variant and removing visual evidence leads to a clear degradation. VALOR further requires coordinated audio-visual grounding, where combining A and V outperforms either modality alone, and removing either audio or visual evidence weakens performance. These observations suggest that different QA require different evidence sources, making fixed modality importance insufficient.

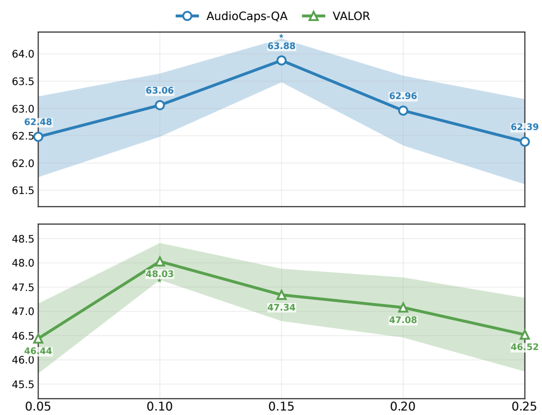
The leave-one-out results further show that no modality is redundant when it is available.

Structural evidence provides cross-evidence constraints, textual evidence supplies semantic context, and audio/visual evidence contributes task-specific perceptual cues. Therefore, the advantage of NuGAT comes from its ability to ground heterogeneous evidence through explicit evidence-anchor correspondences and fuse it with utility-guided subspace retention. This enables NuGAT to exploit complementary evidence while reducing the risk of modality dominance in MLLM-based answer generation.

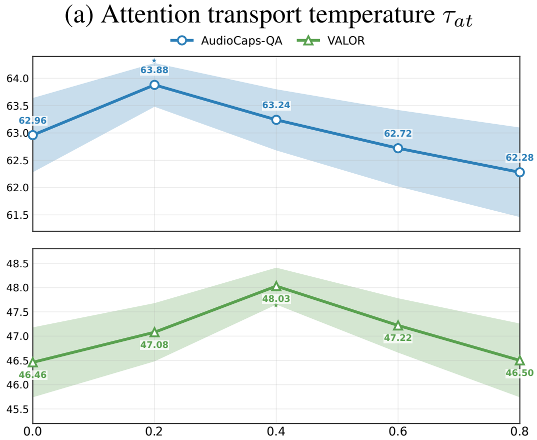
4.5.5. Hyperparameter Sensitivity



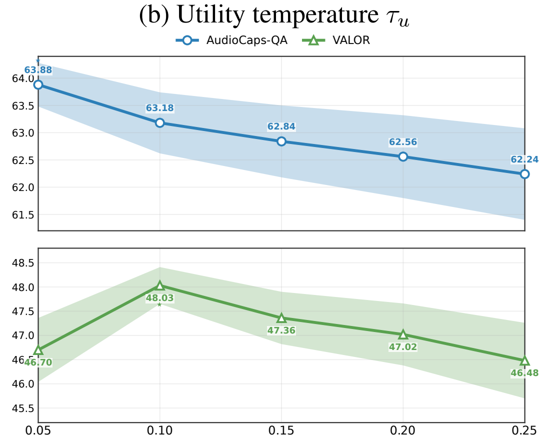
(a) Attention transport temperature τ_{at}



(b) Utility temperature τ_u



(c) Residual fusion coefficient λ



(d) Subspace mask temperature τ_g

Figure 6: Hyperparameter sensitivity analysis of NuGAT on QA. We use Qwen2.5-Omni as the fixed MLLM answerer and report Model-as-Judge (M.J.) scores. The two curves correspond to AudioCaps-QA and VALOR.

We analyze four key hyperparameters of NuGAT under the fixed-answerer QA setting: the attention transport temperature τ_{at}, the utility temperature τ_u, the residual fusion coefficient λ, and the subspace mask temperature τ_g. As shown in Fig. 6, we use Qwen2.5-Omni as the fixed answerer and report M.J. scores on AudioCaps-QA and VALOR, representing audio-dominant and audio-visual QA scenarios, respectively. Separate y-axis ranges are used for readability.

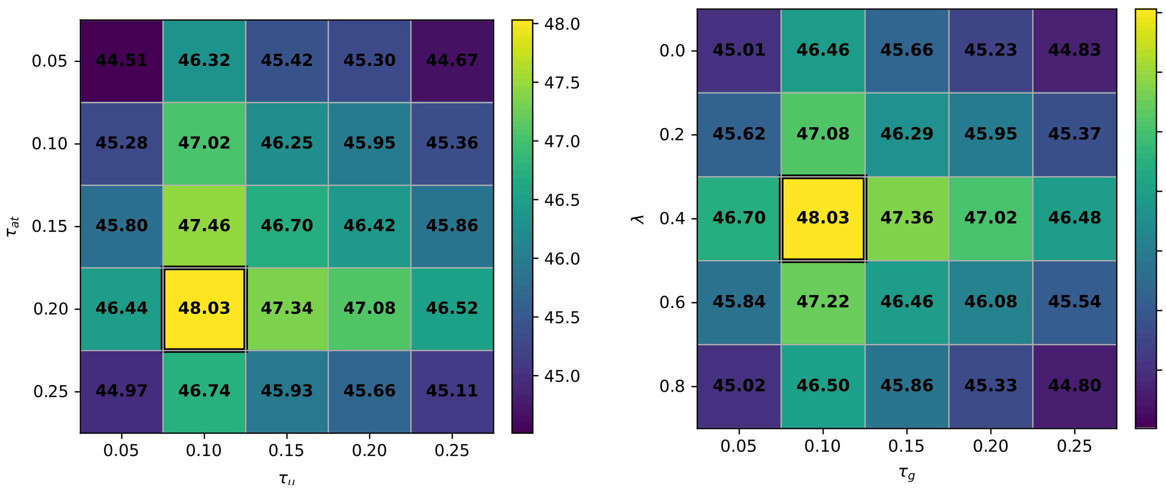
The two benchmarks prefer different hyperparameter regions, reflecting their different evidence requirements. On AudioCaps-QA, performance peaks with sharper evidence transport and subspace selection, $\tau_{at} = 0.10$, $\tau_u = 0.15$, $\lambda = 0.20$, and $\tau_g = 0.05$. This suggests that audio-dominant QA benefits from focused evidence-anchor matching and selective retention of reliable audio-related directions. In contrast, VALOR achieves the best performance at $\tau_{at} = 0.20$, $\tau_u = 0.10$, $\lambda = 0.40$, and $\tau_g = 0.10$, indicating that audio-visual QA requires smoother evidence transport and stronger residual integration to coordinate heterogeneous audio and visual cues.

Overall, the sensitivity trends show that these hyperparameters control complementary aspects of evidence grounding and fusion. The transport temperature τ_{at} determines the sharpness of evidence-anchor matching, τ_u affects utility-based reliability estimation, λ controls the preservation of previous grounded states, and τ_g regulates subspace retention. The stable performance peaks around the selected settings suggest that NuGAT does not rely on a single sensitive parameter, but benefits from coordinated control over structured grounding and utility-guided fusion.

4.5.6. Pairwise Hyperparameter Sensitivity

The one-dimensional sensitivity analysis evaluates each hyperparameter separately, but it cannot reveal how coupled hyperparameters jointly affect evidence grounding and fusion. Therefore, Fig. 7 reports two-dimensional sensitivity heatmaps on VALOR with Qwen2.5-Omni as the fixed answerer. VALOR is a representative setting because audio–visual QA requires coordinated grounding and fusion of audio, visual, textual, and structural evidence.

The $\tau_{at} \times \tau_u$ heatmap reveals a complementary relation between evidence-anchor matching and utility-based reliability estimation. The best result is obtained with a relatively smooth transport temperature $\tau_{at} = 0.20$ and a sharper utility temperature $\tau_u = 0.10$. This suggests that audio–visual evidence should not be assigned too aggressively during early transport, because useful cues may be distributed across modalities. In contrast, utility estimation should remain discriminative after transport, so that noisy or weakly grounded modalities can be suppressed. When both temperatures are too small, the model tends to make premature hard assignments; when both are too large, both transport and utility signals become overly smooth and lose modality-specific discrimination. Thus, transport benefits from moderate exploration, whereas



(a) VALOR: $\tau_{at} \times \tau_u$ (b) VALOR: $\lambda \times \tau_g$

Figure 7: Two-dimensional hyperparameter sensitivity of NuGAT on VALOR under the fixed Qwen2.5-Omni answerer. The left and right heatmaps vary (τ_{at}, τ_u) and (λ, τ_g), respectively. Highlighted cells denote the best.

utility fusion benefits from sharper reliability estimation.

The $\lambda \times \tau_g$ heatmap shows that residual integration and subspace retention also need to be coordinated. The best region appears with a moderate residual coefficient and a relatively sharp subspace mask. This indicates that the fused state should preserve useful evidence accumulated from previous layers while still allowing newly grounded multimodal evidence to update the representation. If λ is too small, useful shared evidence may be overwritten; if it is too large, the model becomes conservative and underuses current evidence. Meanwhile, an appropriate τ_g removes noisy modality-specific directions without discarding informative subspaces.

Overall, the heatmaps show that effective grounded multimodal fusion depends on coordinated control of evidence-anchor matching, modality reliability estimation, residual evidence preservation, and utility-guided subspace retention.

4.6. Time Complexity Analysis

This section analyzes the efficiency of NuGAT as a local multimodal evidence grounding and fusion module. NuGAT does not aim to replace external evidence acquisition or evidence-source construction. Instead, it operates on the current multimodal evidence context and adaptively grounds and fuses the available evidence units. Therefore, its complexity scales with the local evidence context rather than the size of the entire external evidence source. Let $N = \sum_{m \in \mathcal{M}} n_m$ denote the total number of modality-specific evidence units, n denote the

Table 4: Theoretical comparison of query-time evidence construction and evidence adaptation costs. K denotes the number of candidate evidence items, G_q denotes the query-related evidence graph, $\mathcal{I}_{idx}$ denotes indexed evidence access cost, and $\mathcal{C}_{check}$ or $\mathcal{C}_{prune}$ denotes additional checking or pruning cost.

Method	Main operation	Query-time cost	Adaptive grounding/fusion
Wikidata	Text-KG evidence access	$O(\mathcal{I}_{idx} + Kd)$	No
VTKG	Image-text evidence access	$O(\mathcal{I}_{idx} + Kd)$	No
M^{2}ConceptBase	Concept-level evidence matching	$O(\mathcal{I}_{idx} + Kd)$	No
VAT-KG	Visual-audio-text evidence checking	$O(\mathcal{I}_{idx} + Kd + \mathcal{C}_{check})$	Partial
M^{3}KG-RAG	Multi-hop evidence expansion and pruning	$O(\mathcal{I}_{idx} + \lvert G_q\rvert d + \mathcal{C}_{prune})$	Query-graph level
NuGAT	AT grounding and NUF fusion	$O(L(Nnd + \lvert\mathcal{M}\rvert nd\min(n,d)))$	Evidence-anchor/subspace level

number of evidence anchors, d denote the hidden dimension, $|\mathcal{M}|$ denote the number of modalities, and L denote the number of fusion steps.

For each fusion step, Attention Transport computes evidence-anchor matching and transports modality-specific values into the evidence-anchor space. The dominant grounding cost is

$$O(Nnd), \tag{21}$$

where Nn corresponds to local evidence-anchor interactions. Since the transport plan is obtained by closed-form column-wise softmax, NuGAT avoids the iteration-dependent cost of Sinkhorn-style OT solvers.

Nash Utility Fusion performs modality-wise subspace retention over the grounded anchor representations. Its SVD-based retention cost is

$$O\big(|\mathcal{M}|nd\min(n,d)\big). \tag{22}$$

If anchor-level utility estimation is implemented with full anchor-to-anchor contrastive scoring, an additional $O(|\mathcal{M}|n^2d)$ term is introduced. In our setting, n is the number of local evidence anchors and is typically no larger than d, so this term is of the same order as the SVD-based retention cost. Thus, the overall inference complexity can be written as

$$O\left(L\left(Nnd + |\mathcal{M}|nd\min(n,d)\right)\right), \tag{23}$$

and the memory cost is

$$O\left(Nn + |\mathcal{M}|nd\right), \tag{24}$$

where Nn stores the local evidence-anchor transport plans and $|\mathcal{M}|nd$ stores grounded modal-

Table 5: Empirical efficiency comparison on VALOR with Qwen2.5-Omni as the fixed answerer. M^{3}KG-RAG is selected because it is the strongest baseline in the main QA comparison.

Method	GPU VRAM (GB)	Avg. time/query (s)	M.J. (%)
w/o AT	24.1	4.83	45.85
w/o NUF	24.3	4.92	46.52
NuGAT	24.8	5.05	48.03
M^{3}KG-RAG [28]	39.8	7.02	44.67

ity representations.

Table 4 clarifies the difference between NuGAT and the compared evidence-construction baselines. Wikidata, VTKG, and M^{2}ConceptBase mainly provide external evidence through indexed access or concept-level matching. VAT-KG further performs multimodal evidence checking, while M^{3}KG-RAG introduces multi-hop evidence expansion and pruning over a query-related evidence graph. These methods mainly differ in how they prepare or filter evidence before answer generation. In contrast, NuGAT focuses on the local adaptation stage: given the current evidence context, it explicitly grounds evidence units to evidence anchors and performs utility-guided subspace retention before fusion. Therefore, the comparison is not intended to claim that all methods implement the same module, but to show where their query-time computation is spent and why NuGAT scales with local evidence units and anchors.

Table 5 reports empirical efficiency on VALOR. We compare with M^{3}KG-RAG rather than all baselines for three reasons. First, M^{3}KG-RAG is the strongest baseline in the main QA comparison, so it provides the tightest efficiency-performance reference. Second, weaker baselines may have lower or different query-time costs, but they do not offer a stronger effectiveness comparison; including them would not provide a more meaningful trade-off analysis. Third, the internal variants w/o AT and w/o NUF already isolate the additional cost and benefit of the two proposed components.

The internal variants show that removing AT or NUF slightly reduces inference cost, but both variants obtain lower M.J. scores. This indicates that the additional local grounding and fusion computation contributes directly to evidence quality. Compared with M^{3}KG-RAG, NuGAT improves the M.J. score by 3.36, while reducing GPU memory usage by 37.7% and average inference time by 28.1%. These results suggest that NuGAT achieves a better efficiency-performance trade-off by performing local structured grounding and utility-guided subspace retention over the current evidence context, rather than relying on heavier query-time evidence expansion and pruning.

5. Conclusion

In this paper, we proposed NuGAT, a general evidence-grounded multimodal alignment and fusion framework for adaptive evidence grounding and fusion. NuGAT addresses fragmented evidence grounding by formulating heterogeneous evidence integration as structured evidence-anchor correspondence learning. Specifically, Attention Transport transports modality-specific evidence units to evidence anchors through attention-guided and structure-aware transport, producing explicit and interpretable evidence-anchor correspondences. To mitigate modality dominance after grounding, Nash Utility Fusion treats each modality as a utility-bearing modality and uses utility gain to control modality-specific subspace retention rather than direct scalar weighting. In this way, NuGAT preserves reliable evidence directions while suppressing noisy or weakly grounded signals. Experiments on multimodal QA and concept verification demonstrate that NuGAT consistently improves evidence-grounded reasoning and verification performance over strong baselines. Ablation studies verify the complementary contributions of Attention Transport and Nash Utility Fusion, while modality and sensitivity analyses show that NuGAT adaptively integrates heterogeneous evidence across different scenarios.

6. Limitations

Although NuGAT achieves promising results, several aspects remain to be further explored. First, the current implementation relies on pre-extracted modality features and constructed evidence units. When the available evidence is incomplete, weakly structured, or represented only by global modality embeddings, the learned evidence-anchor correspondences may become less fine-grained and less interpretable. Second, the structure-aware compatibility prior depends on entity correspondence and offline structural compatibility estimation. If these priors are incomplete or noisy, Attention Transport may receive weaker grounding guidance and rely more heavily on feature-level attention. Third, Nash Utility Fusion defines modality utility according to the alignment quality between grounded modality representations and the fused evidence representation. Although this definition is effective for mitigating modality dominance, alternative utility designs, such as uncertainty-aware, retrieval-oriented, or task-specific reliability measures, may further improve fusion under different application scenarios. Finally, our current evaluation mainly covers multimodal QA and concept verification. Future work will examine the generality of NuGAT in broader settings, such as large-scale multimodal retrieval-augmented generation and open-world evidence-grounded reasoning.

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